

Creating Snacktive					
6	Identifying and Preparing Equipment	1. Gather all equipment in accordance with Criterion B Strand 4. In this case, a large assortment of glass bowls, spatulas, the ingredients, the blenders, etc. 2. Ensure all this equipment has been thoroughly washed, preferably using a dishwasher at the highest heat settings (<i>or permitted for the equipment</i>). This ensures all tools are sterilized.	- Kitchen Utensils - Notebook - Pen	Safety Materials	30 mins
7	Preparing Raw Ingredients	1. With the gathered ingredients, follow the steps detailed in Criterion B Strand 4 to prepare these raw ingredients. 2. In this case, soak and boil the black beans until mushy. 3. Boil quinoa according to package instructions.	- Kitchen Utensils - Weighing Scale - Measuring Cups	Safety Materials Nutrition Flavour	30 mins
8	Mixture Preparation + Baking	1. Following the plan detailed in Criterion B Strand 4, and referring to the step-by-step recipe in Appendix B1, complete all the steps. 2. Bake until crisp, dry and golden-brown. 3. Once out of the oven, immediately cover with the prepared seasoning mix to ensure it sticks.	- Kitchen Utensils - Oven	Safety Materials Nutrition Flavour	120 mins
Testing Snacktive					
9	Taste Test	1. Allow the freshly baked snack to cool before attempting to consume. 2. Taste every seasoning mix with the cracker and identify areas of strengths and limitations. 3. After 24 hours, provide this snack to members of the target audience in order to further assess aspects of the snack. This will help determine its success compared to the success criteria.	- Snacktive - Consenting Human	Safety Flavour	30 mins
10	Altering the Recipe (if needed)	1. Based on the feedback received, alter the recipe as needed, making notes of the alterations. 2. Repeat the taste test process and determine which iteration yielded better results.	- Kitchen Utensils	Safety Materials Nutrition Flavour	120 mins
Packaging Snacktive					
11	Sourcing the Eco-Friendly Packs	1. Through a local vendor, source the necessary packages. Refer to Appendix B3 to understand the reasons behind this decision. 2. Using a printer, print the created packaging of Snacktive based on Crit B Str 4. 3. Using commercial packaging glue, stick onto the acquired package. 4. Transfer 100g of Snacktive into the pack and seal.	- Package - Product - Paper - Printer - Glue	Materials Safety	30 mins

Strand 2: Developing and demonstrating technical skills

The process of understanding the problem at hand, carrying out extensive research into the solutions available in the market today and applying the gained knowledge to develop a product which is capable of meeting the developed criteria resulted in various skills being developed. These skills happen to be of 3 types - research skills, thinking skills and baking skills - along with some soft skills along the way. The process of understanding the problem at hand,

determining the fundamental needs of the target audience, and evaluating the scope of current market options to solve the problem helped develop research skills. Soon after, in determining the ideal composition of the product, investigating the plethora of ways to produce it, and crucially, developing the most ideal package to attract the target audience helped enhance my thinking skills. While developing the product, various baking skills provided the opportunity to better my baking skills.

Skill Developed	Evidence																																				
Nutritional Research (<i>Research Skills</i>) The process of conducting thorough research into the nutritional requirements of the target audience and building an understanding of the nutritional quantities of various ingredients developed the skill. During my research into the nutritional necessities of the target audience, I showed enhanced critical thinking skills as well by being able to compile the base necessity nutrition, common for both female and male adolescents. It may seem easy to make such a determination by simply analysing data provided on multiple sources, however, in the process of conducting this research, it must be noted that different countries and different sources across the internet provided varying data. In order to make an accurate determination, the skills of information literacy were implied to verify the credibility of each source.	healthline.org Ages 9 to 13: Daily guidelines for girls <table><tr><td>Calories</td><td>1,400 to 2,200, depending on growth and activity level</td></tr><tr><td>Protein</td><td>4 to 6 ounces</td></tr><tr><td>Fruits</td><td>1.5 to 2 cups</td></tr><tr><td>Vegetables</td><td>1.5 to 3 cups</td></tr><tr><td>Grains</td><td>5 to 7 ounces</td></tr><tr><td>Dairy</td><td>2 cups</td></tr></table> Ages 9 to 13: Daily guidelines for boys <table><tr><td>Calories</td><td>1,600 to 2,400, depending on growth and activity level</td></tr><tr><td>Protein</td><td>5 to 6.5 ounces</td></tr><tr><td>Fruits</td><td>1.5 to 2 cups</td></tr><tr><td>Vegetables</td><td>2 to 3.5 cups</td></tr><tr><td>Grains</td><td>5 to 8 ounces</td></tr><tr><td>Dairy</td><td>2 cups</td></tr></table> Ages 14 to 18: Daily guidelines for girls <table><tr><td>Calories</td><td>1,800 to 2,400, depending on growth and activity level</td></tr><tr><td>Protein</td><td>5 to 6.5 ounces</td></tr><tr><td>Fruits</td><td>1.5 to 2 cups</td></tr><tr><td>Vegetables</td><td>2.5 to 3 cups</td></tr><tr><td>Grains</td><td>6 to 8 ounces</td></tr><tr><td>Dairy</td><td>3 cups</td></tr></table> Ages 14 to 18: Daily guidelines for boys healthline.org Healthy Eating During Adolescence What is healthy eating? Eating healthy is an important part of a healthy lifestyle and is something that should be taught at a young age. The following are some general guidelines for helping your adolescent eat healthy. It is important to discuss your adolescent's diet with his or her health care provider before making any dietary changes or placing your adolescent on a diet. Discuss the following healthy eating recommendations with your adolescent to ensure he or she is following a healthy eating plan: • Eat 3 meals a day, with healthy snacks. • Increase fiber in the diet and decrease the use of salt. • Drink water. Try to avoid drinks that are high in sugar. Fruit juice can have a lot of calories, so limit your adolescent's intake. Whole fruit is always a better choice. • Eat balanced meals. • When cooking for your adolescent, try to bake or broil instead of fry. • Make sure your adolescent watches (and decreases, if necessary) his or her sugar intake. • Eat fruit or vegetables for a snack.	Calories	1,400 to 2,200, depending on growth and activity level	Protein	4 to 6 ounces	Fruits	1.5 to 2 cups	Vegetables	1.5 to 3 cups	Grains	5 to 7 ounces	Dairy	2 cups	Calories	1,600 to 2,400, depending on growth and activity level	Protein	5 to 6.5 ounces	Fruits	1.5 to 2 cups	Vegetables	2 to 3.5 cups	Grains	5 to 8 ounces	Dairy	2 cups	Calories	1,800 to 2,400, depending on growth and activity level	Protein	5 to 6.5 ounces	Fruits	1.5 to 2 cups	Vegetables	2.5 to 3 cups	Grains	6 to 8 ounces	Dairy	3 cups
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Ingredient Functionality Analysis In the process of developing the recipe for Snacktive, it became of paramount importance to understand the interaction of ingredients with each other to ensure accurate nutritional values. This process was marred with scouring through a plethora of sources, using and enhancing information literacy skills to determine credibility.	healthline.com when flight prices drop 2. Spinach  Spinach provides many health benefits but very few calories. About 3.5 ounces (100 grams) of raw spinach contain 2.7 mg of iron, or 15% of the DV (31%). Although this is non-heme iron, which isn't absorbed very well, spinach is also rich in vitamin C. This is important since vitamin C significantly boosts iron absorption (31%). Spinach is also rich in antioxidants called carotenoids, which may reduce your risk of cancer, decrease inflammation, and protect your eyes from disease (31%, 32%, 33%). Consuming spinach and other leafy greens with fat helps your body absorb the carotenoids, so make sure to eat a healthy fat like olive oil with your spinach (34%). when flight prices drop																																				

Precision Measuring

In the early stages of development, I identified the necessity of accuracy in measuring ingredients was important for ensuring consistency in nutritional content. The procedure required the use of digital scales and calibrated measuring equipment to guarantee precise amounts of flour, beans, and other ingredients were chosen. Initial tests resulting in texture and flavor variations, necessitating weight adjustments of ingredients. With time, frequent measurements improved accuracy and made it convenient to mass-produce Snacktive without altering the quality of each batch.



Dough Handling and Shaping

Developing ideal dough consistency in Snacktive involved repeated hands-on testing. Early inconsistencies in ingredient mixes resulted in too dry and crumbly, or too wet and uncooperative, dough. The first batches were inconsistent, which complicated the shaping of the bars tremendously. However, constant experimentation to ingredient proportions, particularly moisture trapping ingredients such as oat flour and beans improved the ease of pliability.

Crucially, the next skill required to be developed were kneading skills - something which required significant improvement. This skill involved the usage of a rolling pin to press the dough into the desired shape and thickness to create the bar. While this may sound easy, ensuring uniform thickness without cracking or sticking to the surface proved to be challenging. Across many trials, I began to master this technique, rolling small portions of dough into shape to ensure uniform thickness and then moving on to larger portions. A process that helped ease the burden was cooling the dough. Cooler dough tended to be easier to knead and less likely to crack or stick to the counter-top.



Baking and Temperature Control

The process of finally baking the bar was marred by a series of failures, leading to an almost never ending cycle of trial and error. The initial batches were more overcooked, being excessively hard, possessing depleted nutrition and having a bitter taste due to burnt edges. The next few batches were underbaked, overly soft and crumbly - not ideal for prolonged shelf storage.

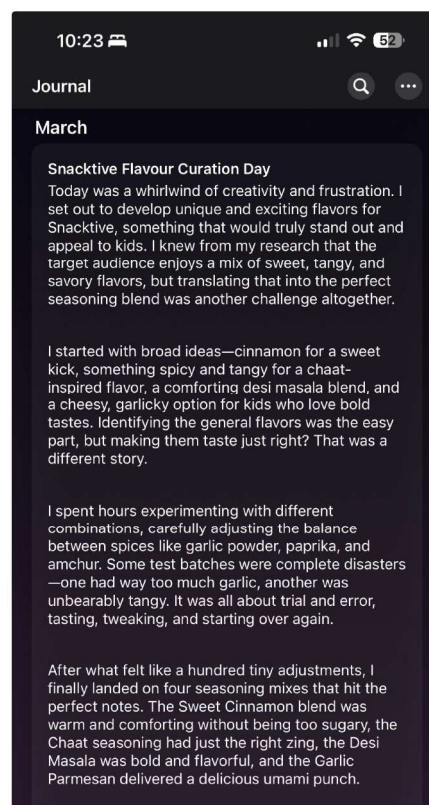
In order to craft the best possible cracker texture, balancing the need to maintain nutrition and maximise shelf life, a series of trials involving changes in temperature and baking times were carried out. I carefully observed the progress of



Snacktive at temperatures ranging between 180-220°C. Additionally, various baking times were tried out, between 25 and 40 minutes to identify the ideal baking temperature and time range. During this process, I identified aroma, surface texture and color were key indicators of doneness. These indicators were essential to the identification of correct temperature and time ranges, thus enabling me to ensure Snacktive was crispy and crunchy, while also maximising shelf-life by reducing moisture.

Creating Innovative Flavours

Using creative and critical thinking skills, I intended to develop a host of new and innovative flavour options which could help attract the target audience. In this deep thought process, I undertook essential research in order to comprehend the general taste preferences of the group, and then curate flavours to meet the requirements. While identifying the general flavours was easy, developing seasoning mixes which hit the right notes was hard. This process came down to trial and error, experimenting with different ratios of seasonings like garlic powder or paprika to ensure just the right mouth feel. In the end, after multiple test batches, the correct formulation of 4 flavours was developed, helping Snacktive appeal to a broader range of consumers.



Apple Journal App Entry on the day of flavour curation.

Strand 3: Follow the plan to create the solution

No.	Resources	Process	Outcome
1	<u>Intended</u> - Laptop - Oven - Flour - Water - Salt - Baking Powder <u>Used</u> All	<u>Intended</u> 1. Identify the model of the oven available or intended for use. Download and thoroughly analyse the user manual of the oven readily available online. 2. Using the oven, test out how to enable each function and the various settings provided within each mode. 3. To test out the functioning of each mode, make simple baked vegetables in the oven, testing out the convection heating and broiler. <u>Completed</u> All processes completed Under the time limit	